

Performance Testing

Date	7 July 2026
Team ID	SWTID-2026-8075
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Streamlet, Python Time Module
Type of Testing	Load Testing, Functional Performance Testing
Target Module / API	Speech-to-Text, Semantic Similarity Analysis, Audio Processing, PDF Report Generation
Test Environment	Local System (Windows, Python, Streamlet)

Test Date	12 July 2026
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Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Upload Audio & Speech-to-Text	1	10	Audio should be transcribed successfully
2	Semantic Similarity Analysis	1	5	Understanding score should be generated correctly
3	Fluency & Audio Analysis	1	8	WPM, Pause Ratio and RMS Energy should be calculated
4	PDF Report Generation	1	5	PDF report should be generated and downloaded successfully



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.6 Second	Pass	Average response time is within the target limit.
2	Response Time (Max)	< 5 seconds	3.8 Second	Pass	Maximum response time remained acceptable.

3	Throughput (Req/sec)	> 1 Req/sec	2 Req/sec	Pass	System handled requests efficiently.
4	Error Rate	< 1%	0%	Pass	No errors occurred during testing.
5	CPU Utilization	< 80%	45%	Pass	CPU usage remained stable throughout execution.
6	Memory Utilization	< 80%	58%	Pass	Memory consumption was within acceptable limits.

Step 4: Observations & Analysis

Key Findings:
[Describe what the test results revealed about your system's performance]

Bottlenecks Identified:
[List any performance bottlenecks observed during testing]

Optimization Steps Taken:
[Describe any changes or optimizations made to improve performance]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

SCREENSHOT 1:- Home Page

The screenshot shows a web browser window with the URL `localhost:8501`. The page title is "Voice-Based Concept Understanding Analyser". Below the title, a subtitle reads: "This AI-powered application evaluates how well you understand a concept by analyzing your spoken explanation." The page features a "Features" section with a list of capabilities: Speech-to-Text using Whisper, Semantic Similarity Analysis, Audio Fluency Analysis, and PDF Report Generation. A "Select a Concept" dropdown menu is set to "Machine Learning". Below this, a blue bar indicates "Supported formats: WAV, MP3, M4A, MP4, OGG, OPUS, FLAC (Max 200 MB)". An "Upload your Audio Explanation" section contains an "Upload" button and a note: "200MB per file • WAV, MP3, M4A, MP4, OGG, OPUS, FLAC". The Windows taskbar at the bottom shows the search bar, task view, and several open applications including Chrome, Word, and Teams. The system tray on the right displays the date and time as 3:19 PM on 7/13/2026.

SCREENSHOT 2:- Successfully pdf report generated

The screenshot displays a PDF document titled "Voice-Based Concept Understanding Analyser" in a web browser window. The PDF content includes a "Transcript" section, a "Similarity Analysis" section with a "Similarity Score: 0.85" and "Understanding Level: Good", a "Fluency Analysis" section with "Questions: 12/20 answered", "Words Spoken: 120", "Speaking Speed: 120.21 wpm", and "Fluency: Good", a "Pause Analysis" section with "Total Pause Time: 2.14 seconds", "Pause Ratio: 18.25%", and "Pause Quality: Moderate", a "Voice Energy Analysis" section with "Average Word Energy: 0.857" and "Energy Level: Moderate Energy", and a "Filter Word Analysis" section with "Total Filter Words: 1" and "Filter Quality: Good". An "AI Performance Summary" section states: "The explanation shows a good understanding of the concept, covering most of the requested ideas. The speaker's speech is clear and easy to follow. A moderate number of pauses were detected, but the speech remained understandable. The voice energy was moderate throughout the explanation. Only a few filter words were used, which did not significantly affect comprehension." Below the text is an "Audio Waveform" plot showing amplitude over time. The Windows taskbar at the bottom shows the search bar, task view, and several open applications including Chrome, Word, and Teams. The system tray on the right displays the date and time as 3:23 PM on 7/13/2026.